RTOS Benchmark on STM32H750B-DK

Phase 1 - DWT-only Publishable Campaign

Four-test latency comparison of ChibiOS RT 7.0.6, FreeRTOS V11.3.0 and Zephyr 4.4.0 running at 480 MHz (VOS0, FLASH WS=4, I-Cache + D-Cache enabled) on the STM32H750B-DK Discovery Kit.

Author: Edoardo Lombardi

Company: Chibilogic s.r.l.

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Abstract

This report documents the Phase 1 DWT-only publishable benchmark campaign of three open-source real-time operating systems on a single, fully neutral STM32H750B-DK target. The three kernels are compared under identical hardware conditions: CPU clock 480 MHz, voltage scaling VOS0, FLASH_ACR = 0x34, I-Cache + D-Cache enabled, identical GCC 14.2 toolchain and -O2 -fomit-frame-pointer -falign-functions=16 effective flags. Two publishable profiles are reported: **fair_perf** (tickless OFF, WFI OFF) and **realistic_tickless** (tickless ON, WFI ON). Latencies for the four tests T1-T4 are measured inside the firmware via the Cortex-M7 DWT cycle counter. Each (RTOS, profile) combination has been validated against five independent firmware loads, with publication-gate verification of clock, cache, flash, memory placement and priority-inheritance correctness. All required run01..run05 captures passed the publication gate. The run_spread metric is zero cycles on every (RTOS, test) cell, demonstrating fully reproducible results across reloads. In this benchmark suite, under the tested configuration, ChibiOS showed the lowest median latency in all four tests, in both profiles (T1 median 298 cycles vs FreeRTOS 443 and Zephyr 589 in fair_perf). FreeRTOS T1 latency approximately doubles under realistic_tickless (median 845 cycles, +402 vs fair_perf) due to tickless wake-up reconfiguration overhead. Mutex priority-inheritance is correct in all three kernels with 500/500 events verified per RTOS per profile. FreeRTOS and Zephyr are excellent RTOS projects with different design goals; this comparison focuses only on real-time latency under the tested conditions.

Test environment

Hardware

Property	Value
Board	STM32H750B-DK Discovery Kit
MCU	STM32H750XBH6, silicon revision V
Core	Cortex-M7 with double-precision FPU
External oscillator (HSE)	25 MHz crystal
CPU clock	480 MHz (PLL1 from HSE)
Voltage scaling	VOS0 (required for 480 MHz)
FLASH_ACR	0x00000034 (LATENCY = 4 WS, programming delay 2)
I-Cache / D-Cache	ON (production-realistic)
Internal Flash	128 KB (no bootloader, application fits)
RAM	1 MB (64K ITCM + 128K DTCM + 864K AXI/AHB SRAM)
Debug / VCP	ST-LINK V3E onboard, USART3 PB10/PB11, 115200 8N1

Toolchain and compile flags

Tool	Version / Setting
GCC	arm-none-eabi-gcc 14.2.Rel1 (Arm GNU Toolchain)
Make	GNU Make 4.3 (MSYS2)
OpenOCD	0.12.0+dev (xPack)
Optimization	-O2 (no -Os, no -O3, no LTO)
ADR-009 effective flags	-fomit-frame-pointer -falign-functions=16
C standard	C11 application code, C99 portable modules
Cross-RTOS verification	compile_commands.json read-back audit (per port)

RTOS versions

RTOS	Version	Source
ChibiOS RT	7.0.6	branch stable_21.11.x (upstream, no fork)
FreeRTOS Kernel	V11.3.0	tag V11.3.0 (latest stable, March 2026)
Zephyr	4.4.0	tag v4.4.0 (latest stable, April 2026)
STM32CubeH7 HAL	v1.12.1	for FreeRTOS variant only

Clock-tree witness (banner of run 01)

The firmware banner of every run prints back the values of RCC, PWR and FLASH registers immediately after boot, so that the publishable claims of 480 MHz / VOS0 / WS=4 are anchored to actually measured silicon state, not just configuration sources. The excerpt below is taken from *chibios_fair_perf_run01.banner.txt* and is representative.

Register / field	Value
SystemClock	`480000000 Hz`
VOS level	`VOS0`
VOSRDY	`READY`

FLASH_ACR	0x00000034
RCC_PLLCKSELR	
RCC_PLLCFGR	0x01ff0ff9
RCC_PLL1DIVR	0x010e02bf
RCC_D1CFGR	0x00000048
PWR_D3CR	0x0000e000
SCB->CCR	0x00070200
ICache	ON
DCache	ON
Tick rate	1000 Hz
Optimization	-O2 LTO=no

Methodology

The four tests

ID	Name	What is measured
T1	IRQ -> thread latency	TIM2 compare match triggers a hardware-routed IRQ; a high-priority worker thread is unblocked. DWT counter is sampled at ISR entry (A1) and again as soon as the worker thread resumes execution (A4). The published metric is the cycle delta A4 - A1, i.e. the path ISR_ENTRY -> THREAD_RUNNING.
T2	Thread handoff	Two equal-priority threads ping-pong via the native yield / signal primitive. Pure thread-to-thread context-switch cost, with the scheduler hot in cache.
T3	Mutex lock / unlock, uncontended	A single thread locks and immediately unlocks a mutex in a tight loop. No contention. Measures the cost of the fast path through the mutex primitive.
T4	Mutex contended + priority inheritance	Three threads (LOW, MEDIUM, HIGH). LOW owns the mutex, HIGH waits on it, MEDIUM tries to disturb. The scheduler must boost LOW to HIGH's priority and keep MEDIUM excluded for the duration of the critical section. The protocol verifies the PI invariant with 100 one-shot scenarios per run.

Cross-RTOS equivalence. The benchmark defines the same external scenario for all three kernels and implements it with each RTOS's most direct native primitive. The source code is therefore not identical across ChibiOS, FreeRTOS and Zephyr; the comparison is scenario-equivalent under identical hardware, clock, compiler and measurement conditions.

Measurement protocol

All latencies are measured using the Cortex-M7 DWT cycle counter (32-bit free-running, no external timing). DWT overhead is 4 cycles per pair of samples (0.008 us at 480 MHz) and is documented in the firmware banner. T1, T2 and T3 run for 1000 warmup iterations (discarded) followed by 10 000 measured iterations. T4 runs 100 one-shot PI scenarios without warmup (ADR-013 explicit exception, because PI must be observed on the very first event of a fresh kernel state, not after a steady-state warmup). Samples are stored in pre-allocated static arrays placed in AXI-SRAM (memory placement verified per run via the *bench_addr* banner table). No dynamic allocation is used by the benchmark code in any RTOS.

Campaign protocol (ADR-013)

For every (RTOS, profile) combination the firmware is fully rebuilt, flashed and run five independent times (run01 through run05). The hash of the ELF and MAP produced by each rebuild is recorded and verified to be identical across runs of the same target (homogeneity check). The serial collector discards any stale bytes from the ST-Link VCP receive buffer before listening, so that no row from a previous firmware can leak into the current capture. The publication gate rejects a run unless ALL of the following hold: TIM2 setup readback matches the expected state; memory placement banner confirms every benchmark object lives in AXI-SRAM; iteration sequences are dense and ordered; T1/T2/T3 produce exactly 10 000 valid rows; T4 produces exactly 100 PI scenarios with 100/100 passes; ELF and MAP SHA-256 match the campaign lock. The reported median per (RTOS, test) is the median-of-medians across the five runs; jitter is max - min in cycles; run_spread is the cycle span of the per-run medians (a strictly internal stability metric).

On the interpretation of the *max (cyc)* column in the fair_perf profile: a fair_perf run does not suppress asynchronous interrupts (notably the 1 kHz SysTick at 480 MHz), so a single iteration's measured cycle window can capture both the native operation and any asynchronous ISR that happens to land inside the same DWT sample. The primary cross-RTOS hot-path comparison is therefore based on **median, p95 and p99**, while *max* is reported for transparency rather than presented as the pure native operation cost. realistic_tickless suppresses the periodic tick when the system is idle but a CPU-bound loop in any RTOS can still be interrupted by other asynchronous events.

Scope and non-claims (Phase 1)

Phase 1 publishes the DWT cycle-delta A4 - A1 as the headline figure for T1. This metric covers the path from ISR entry to the worker thread regaining execution, **excluding** the silicon hardware-event-to-ISR-entry portion (timer compare match, NVIC arbitration, exception stacking, vector fetch and ISR prologue). The figures in this report must not be quoted as the absolute external IRQ-to-thread latency that an external logic analyzer would observe at GPIO pins; see ADR-015 for the Phase 2 dual-source plan where the analyzer-based metric is introduced alongside the DWT one.

Benchmark conditions

The comparison below is valid only under the exact conditions stated here.

Item	Value
Board / MCU	STM32H750B-DK / STM32H750XBH6 (Cortex-M7, DP-FPU)
CPU clock / cache / flash	480 MHz (VOS0); I-Cache + D-Cache ON; FLASH_ACR = 0x34 (4 wait states)
Compiler / optimization	arm-none-eabi-gcc 14.2.Rel1; -O2 -fomit-frame-pointer -falign-functions=16; no LTO
RTOS versions / source	ChibiOS RT 7.0.6 (branch stable_21.11.x); FreeRTOS V11.3.0 (tag); Zephyr 4.4.0 (tag)
Profile configuration	fair_perf: tickless OFF, WFI OFF. realistic_tickless: tickless ON, WFI ON. No asserts, debug or logging in the measured path.
Measurement method	Cortex-M7 DWT CYCCNT cycle deltas inside the firmware. Phase 1 publishes A4 - A1 (ISR entry -> thread running); the external hardware-event-to-ISR portion is Phase 2 (logic analyzer).
Iterations	1000 warmup (discarded) + 10000 valid per run for T1/T2/T3; 100 one-shot scenarios for T4; 5 validated firmware loads per (RTOS, profile).
Repository	pending public release
Raw logs	pending public archive
Scope	Results apply only to this benchmark configuration.

Draft - final legal review required before external publication. A public repository and a downloadable raw-log archive are not yet available; reproducibility is from source under the stated conditions.

Results - profile fair_perf

Profile semantics: fair_perf (tickless OFF, WFI OFF). All other publication invariants (480 MHz, VOS0, FLASH_ACR=0x34, I-Cache + D-Cache ON, -O2 effective flags, DWT measurement) are unchanged.

Headline aggregate

Median, percentile, jitter and run_spread for each (RTOS, test) cell, computed across 10 000 iterations per run x 5 runs (T1/T2/T3) or 100 scenarios per run x 5 runs (T4). run_spread = 0 cycles on every cell means the median of each individual run was bit-identical to the median of every other run for that (RTOS, test).

RTOS	Test	median (cyc)	median (us)	p95 (cyc)	p99 (cyc)	p99 (us)	max (cyc)	jitter	run_spread
ChibiOS	t1_irq	298	0.621	313	315	0.656	315	35	0
ChibiOS	t2_handoff	97	0.202	97	97	0.202	112	17	0
ChibiOS	t3_mtx_uncont	62	0.129	64	64	0.133	435	373	0
ChibiOS	t4_mtx_pi	207	0.431	207	219	0.456	219	12	0
FreeRTOS	t1_irq	443	0.923	448	448	0.933	458	20	0
FreeRTOS	t2_handoff	372	0.775	384	387	0.806	542	188	0
FreeRTOS	t3_mtx_uncont	207	0.431	211	211	0.440	415	220	0
FreeRTOS	t4_mtx_pi	603	1.256	603	650	1.354	650	47	0
Zephyr	t1_irq	589	1.227	609	610	1.271	623	61	0
Zephyr	t2_handoff	424	0.883	424	424	0.883	621	200	0
Zephyr	t3_mtx_uncont	124	0.258	124	124	0.258	408	284	0
Zephyr	t4_mtx_pi	721	1.502	721	770	1.604	770	49	0

Cross-RTOS comparison

Test	ChibiOS (cyc)	FreeRTOS (cyc)	Zephyr (cyc)	ChibiOS (us)	FreeRTOS (us)	Zephyr (us)	Lowest median (this test)
t1_irq	**298**	443	589	0.621	0.923	1.227	**ChibiOS**
t2_handoff	**97**	372	424	0.202	0.775	0.883	**ChibiOS**
t3_mtx_uncont	**62**	207	124	0.129	0.431	0.258	**ChibiOS**
t4_mtx_pi	**207**	603	721	0.431	1.256	1.502	**ChibiOS**

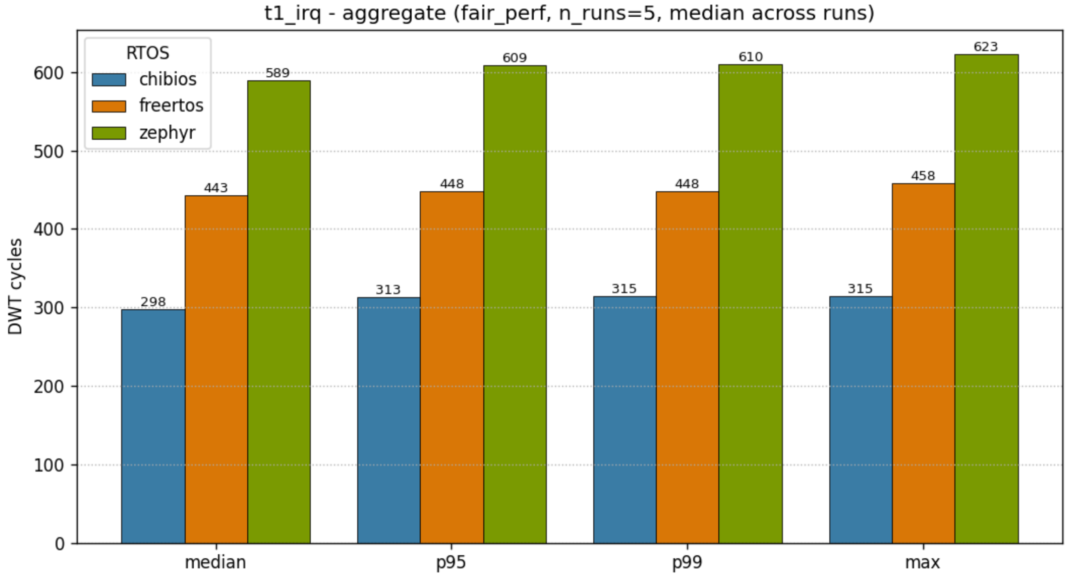
Method: median-of-medians over five validated firmware loads; DWT CYCCNT cycle deltas; lower is better for these latency tests. Results apply only to the benchmark configuration stated in this report.

Priority-inheritance proof (T4)

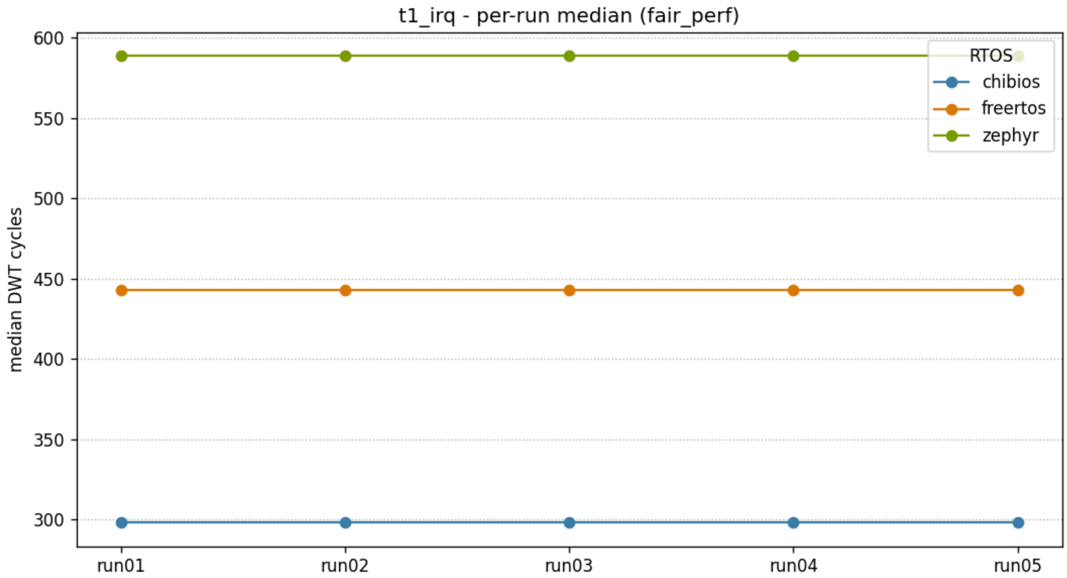
RTOS	PI scenarios	PI passed	Result
ChibiOS	500	500	**PASS**
FreeRTOS	500	500	**PASS**
Zephyr	500	500	**PASS**

Per-test plots

T1 - IRQ -> thread latency

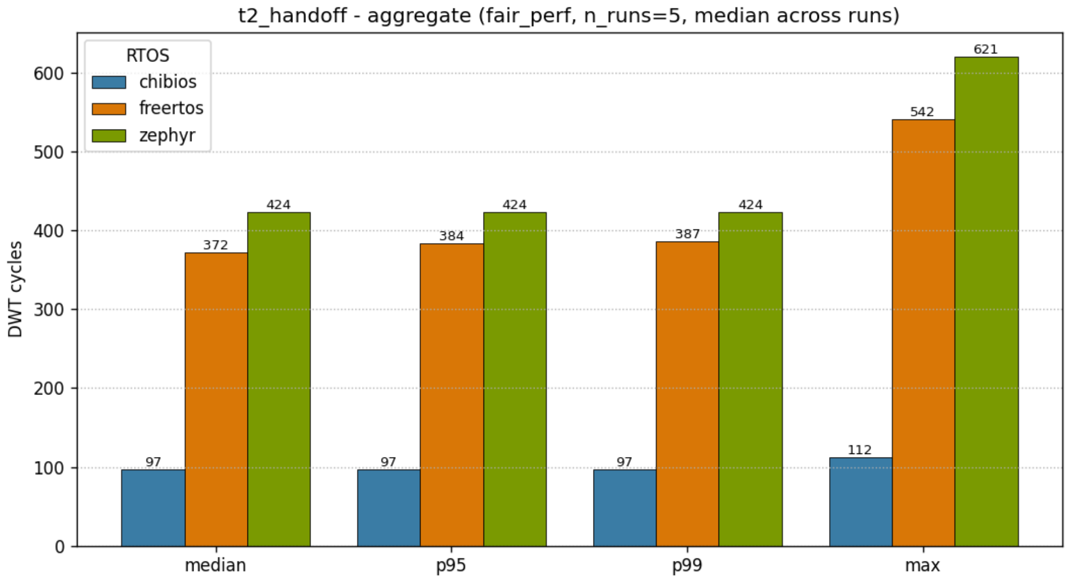


Aggregate distribution - fair_perf / t1_irq

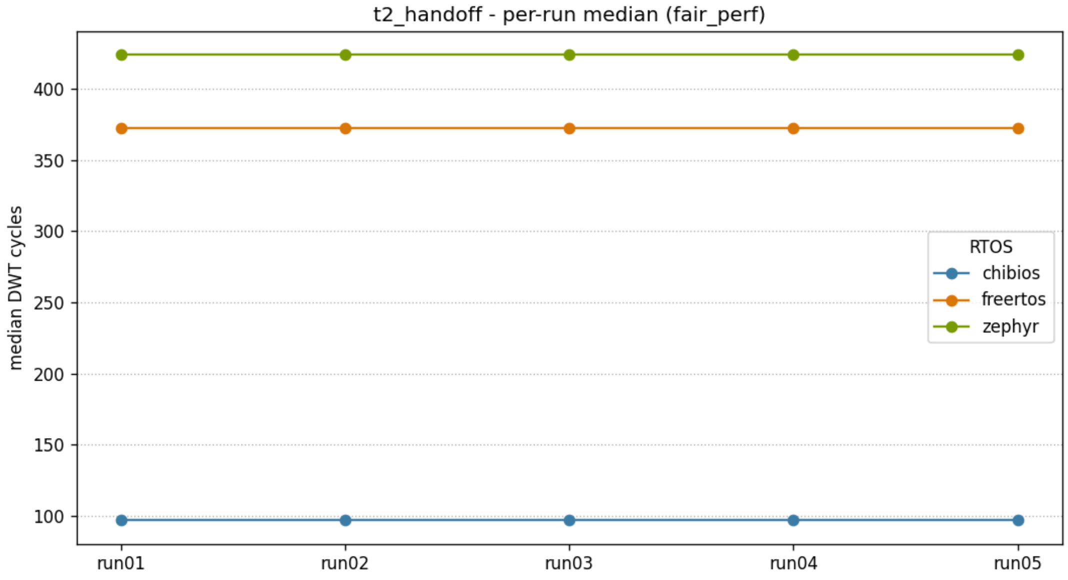


Per-run medians - fair_perf / t1_irq

T2 - Thread handoff

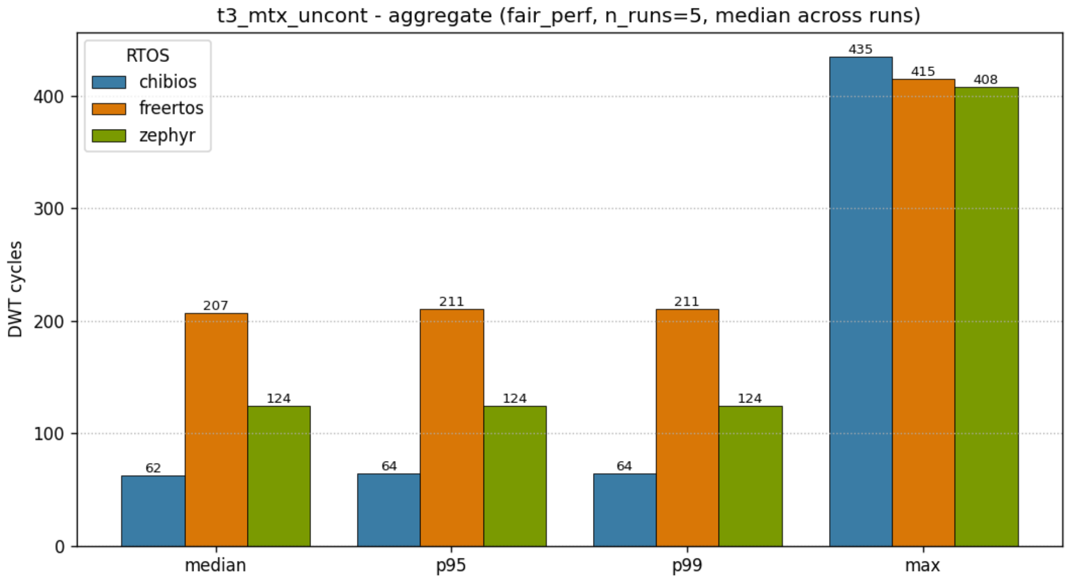


Aggregate distribution - fair_perf / t2_handoff

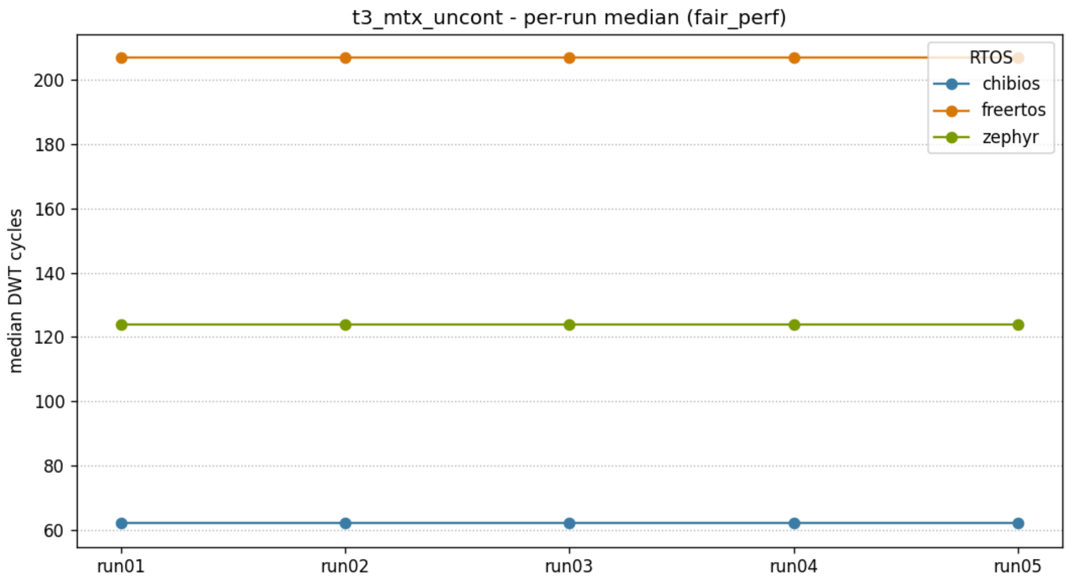


Per-run medians - fair_perf / t2_handoff

T3 - Mutex uncontended

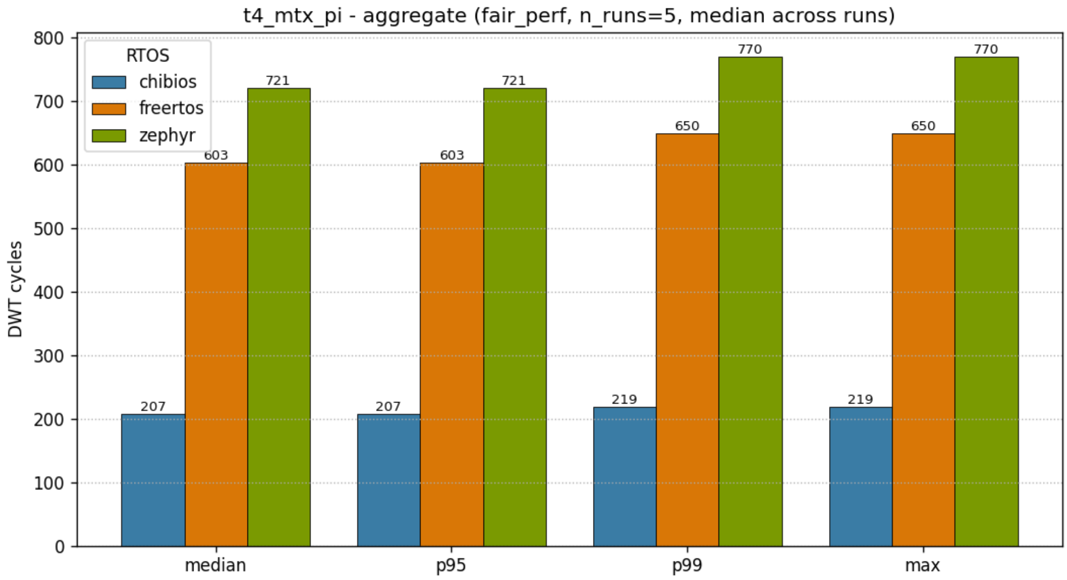


Aggregate distribution - fair_perf / t3_mtx_uncont

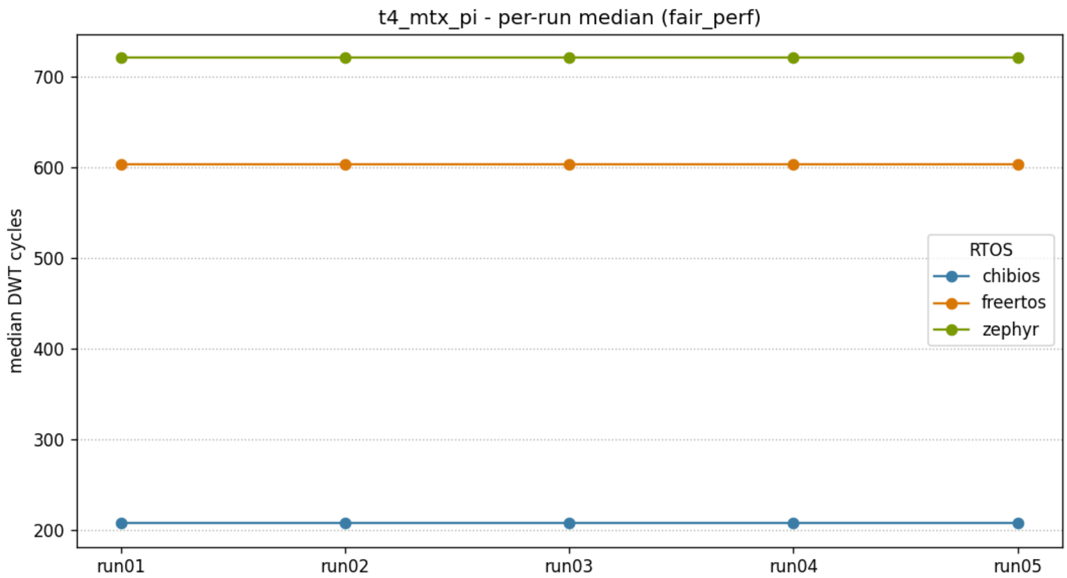


Per-run medians - fair_perf / t3_mtx_uncont

T4 - Mutex contended + priority inheritance

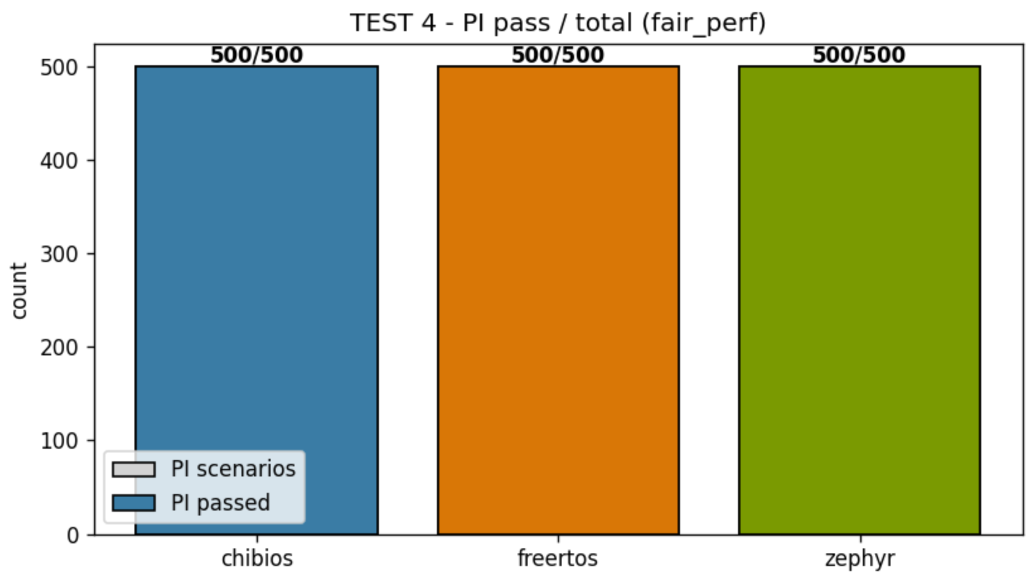


Aggregate distribution - fair_perf / t4_mtx_pi



Per-run medians - fair_perf / t4_mtx_pi

Priority-inheritance event distribution



T4 priority-inheritance scenarios - fair_perf

Results - profile realistic_tickless

Profile semantics: realistic_tickless (tickless ON, WFI ON). All other publication invariants (480 MHz, VOS0, FLASH_ACR=0x34, I-Cache + D-Cache ON, -O2 effective flags, DWT measurement) are unchanged.

Headline aggregate

Median, percentile, jitter and run_spread for each (RTOS, test) cell, computed across 10 000 iterations per run x 5 runs (T1/T2/T3) or 100 scenarios per run x 5 runs (T4). run_spread = 0 cycles on every cell means the median of each individual run was bit-identical to the median of every other run for that (RTOS, test).

RTOS	Test	median (cyc)	median (us)	p95 (cyc)	p99 (cyc)	p99 (us)	max (cyc)	jitter	run_spread
ChibiOS	t1_irq	305	0.635	327	331	0.690	345	57	0
ChibiOS	t2_handoff	93	0.194	94	94	0.196	110	17	0
ChibiOS	t3_mtx_uncont	62	0.129	64	64	0.133	64	2	0
ChibiOS	t4_mtx_pi	204	0.425	204	237	0.494	237	33	0
FreeRTOS	t1_irq	845	1.760	846	851	1.773	1036	192	0
FreeRTOS	t2_handoff	381	0.794	399	419	0.873	553	191	0
FreeRTOS	t3_mtx_uncont	195	0.406	196	196	0.408	371	178	0
FreeRTOS	t4_mtx_pi	613	1.277	613	737	1.535	737	124	0
Zephyr	t1_irq	581	1.210	582	618	1.288	633	58	0
Zephyr	t2_handoff	429	0.894	429	429	0.894	444	18	0
Zephyr	t3_mtx_uncont	125	0.260	125	125	0.260	140	15	0
Zephyr	t4_mtx_pi	682	1.421	691	718	1.496	718	36	0

Cross-RTOS comparison

Test	ChibiOS (cyc)	FreeRTOS (cyc)	Zephyr (cyc)	ChibiOS (us)	FreeRTOS (us)	Zephyr (us)	Lowest median (this test)
t1_irq	**305**	845	581	0.635	1.760	1.210	**ChibiOS**
t2_handoff	**93**	381	429	0.194	0.794	0.894	**ChibiOS**
t3_mtx_uncont	**62**	195	125	0.129	0.406	0.260	**ChibiOS**
t4_mtx_pi	**204**	613	682	0.425	1.277	1.421	**ChibiOS**

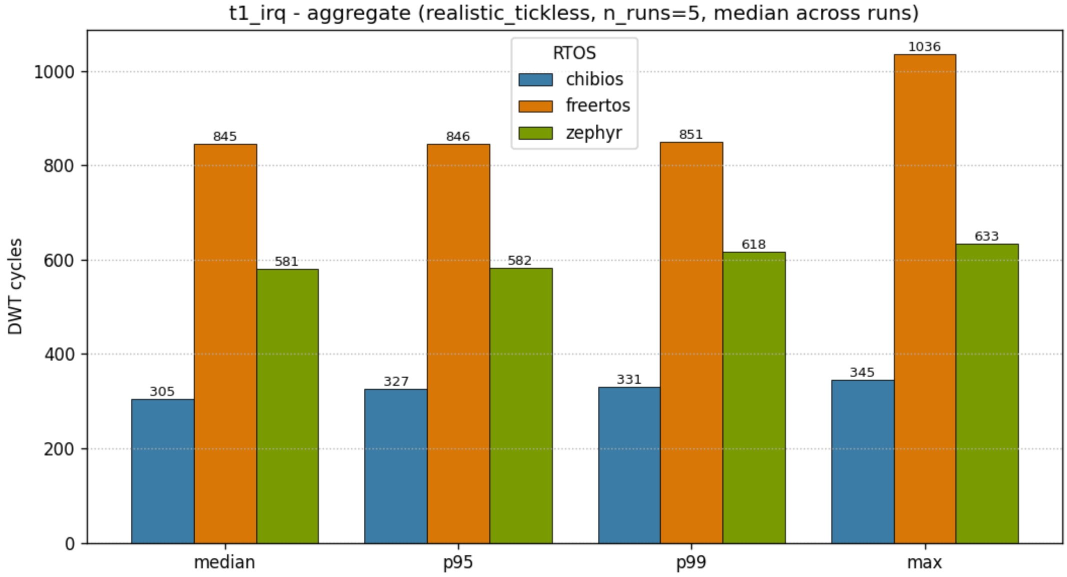
Method: median-of-medians over five validated firmware loads; DWT CYCCNT cycle deltas; lower is better for these latency tests. Results apply only to the benchmark configuration stated in this report.

Priority-inheritance proof (T4)

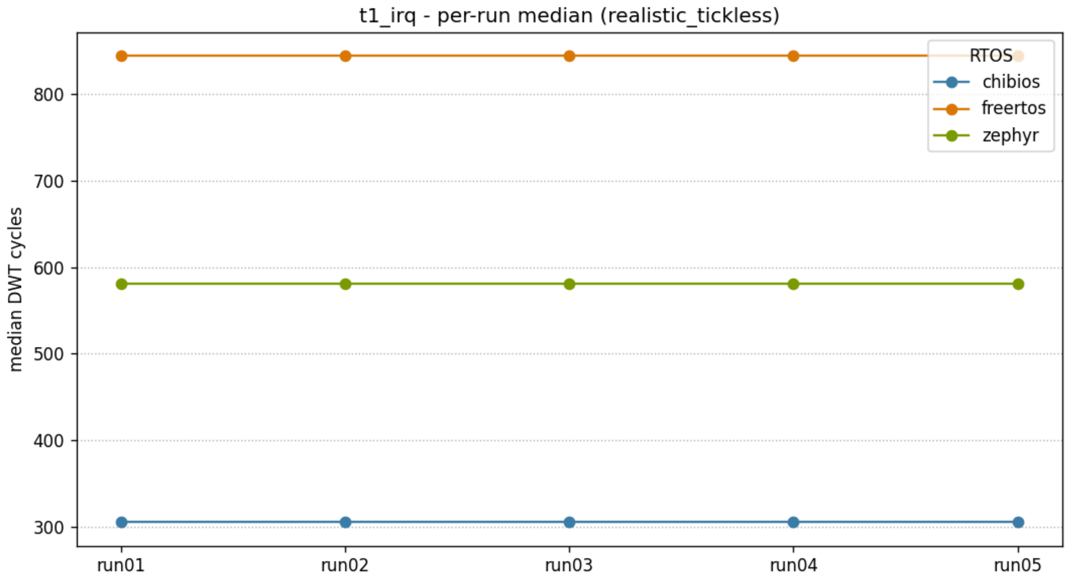
RTOS	PI scenarios	PI passed	Result
ChibiOS	500	500	**PASS**
FreeRTOS	500	500	**PASS**
Zephyr	500	500	**PASS**

Per-test plots

T1 - IRQ -> thread latency

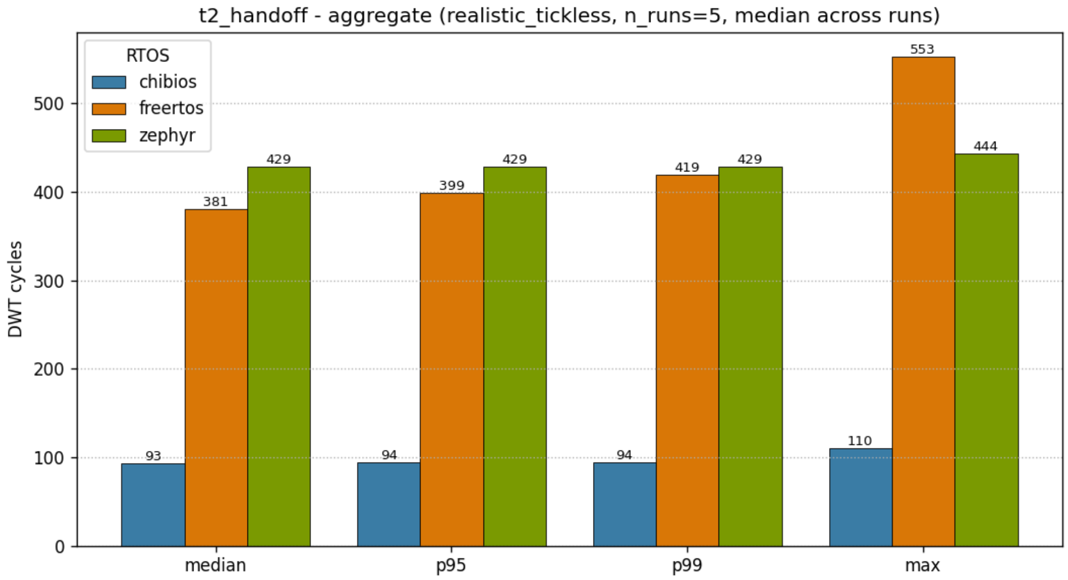


Aggregate distribution - realistic_tickless / t1_irq

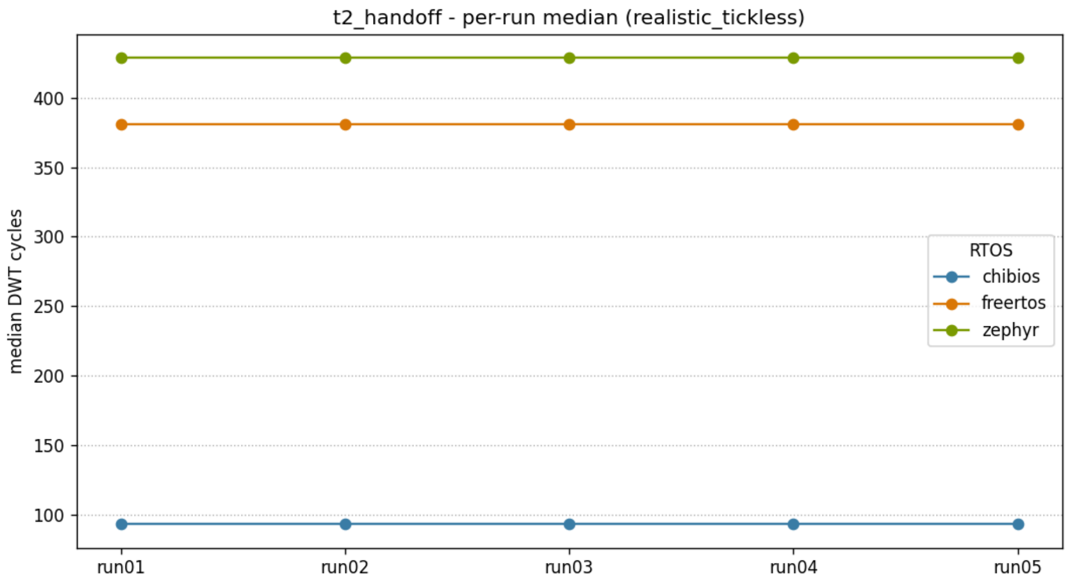


Per-run medians - realistic_tickless / t1_irq

T2 - Thread handoff

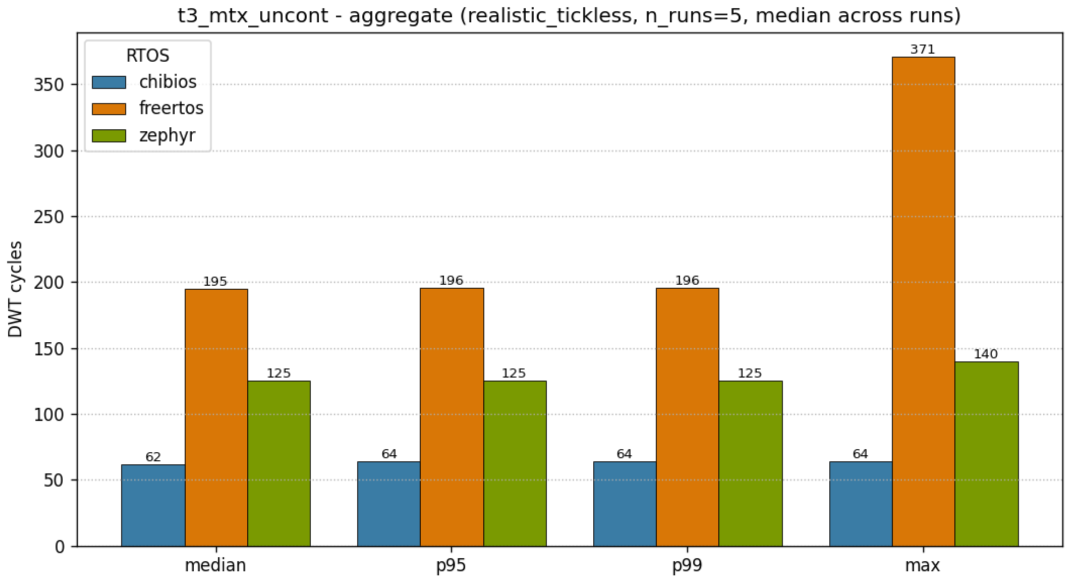


Aggregate distribution - realistic_tickless / t2_handoff

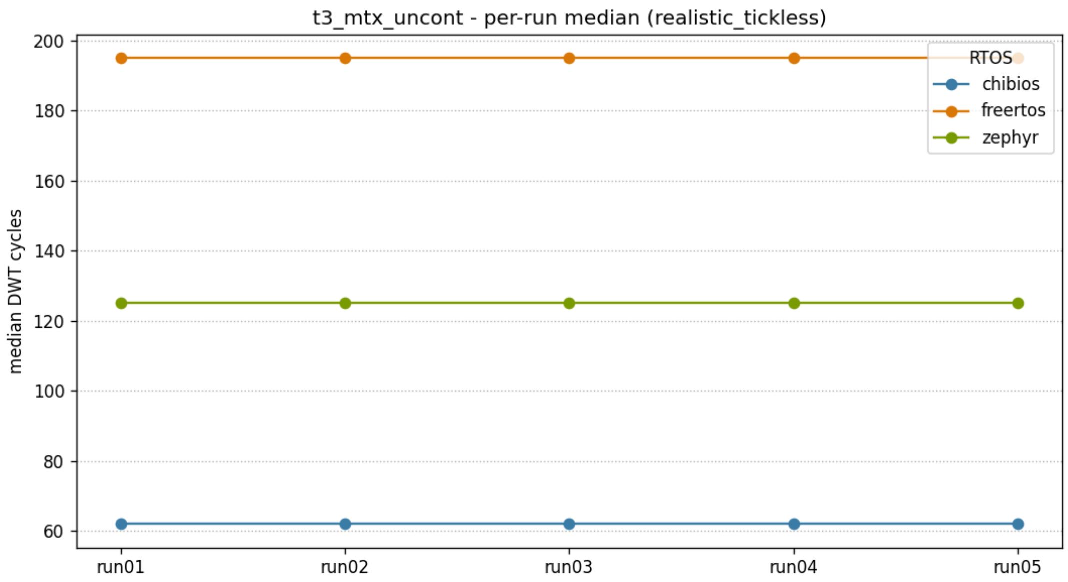


Per-run medians - realistic_tickless / t2_handoff

T3 - Mutex uncontended

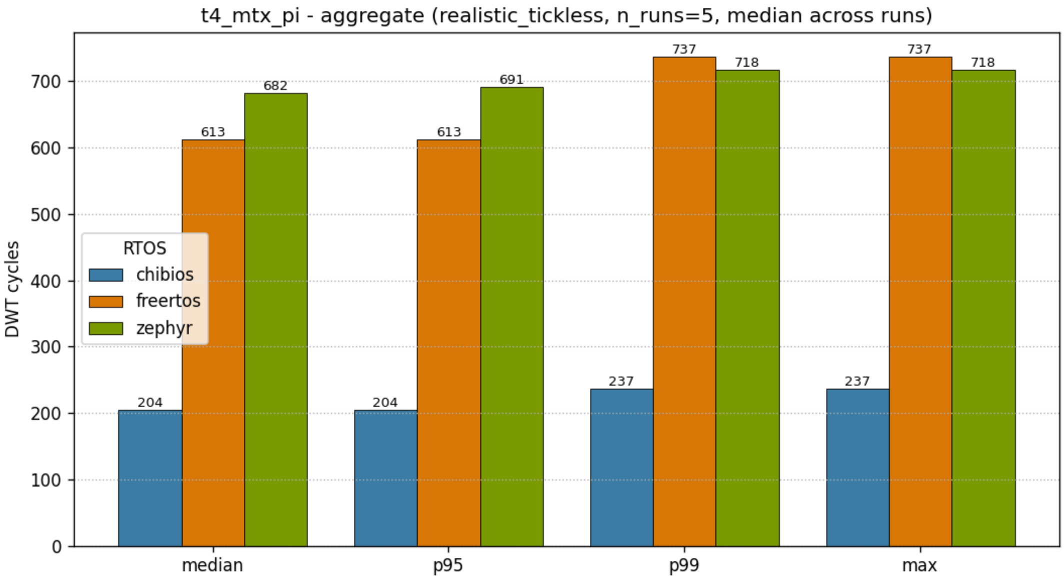


Aggregate distribution - realistic_tickless / t3_mtx_uncont

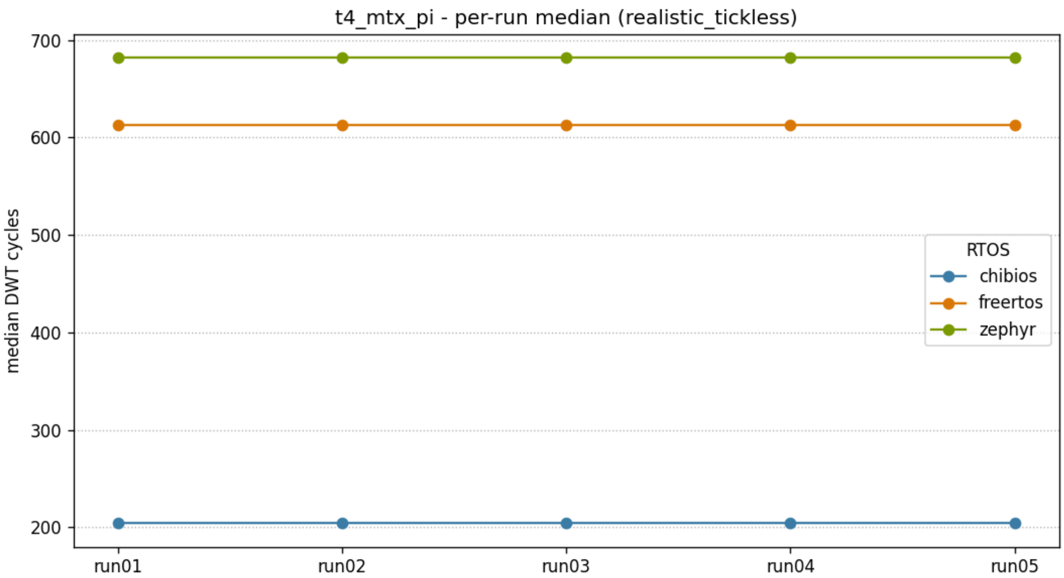


Per-run medians - realistic_tickless / t3_mtx_uncont

T4 - Mutex contended + priority inheritance

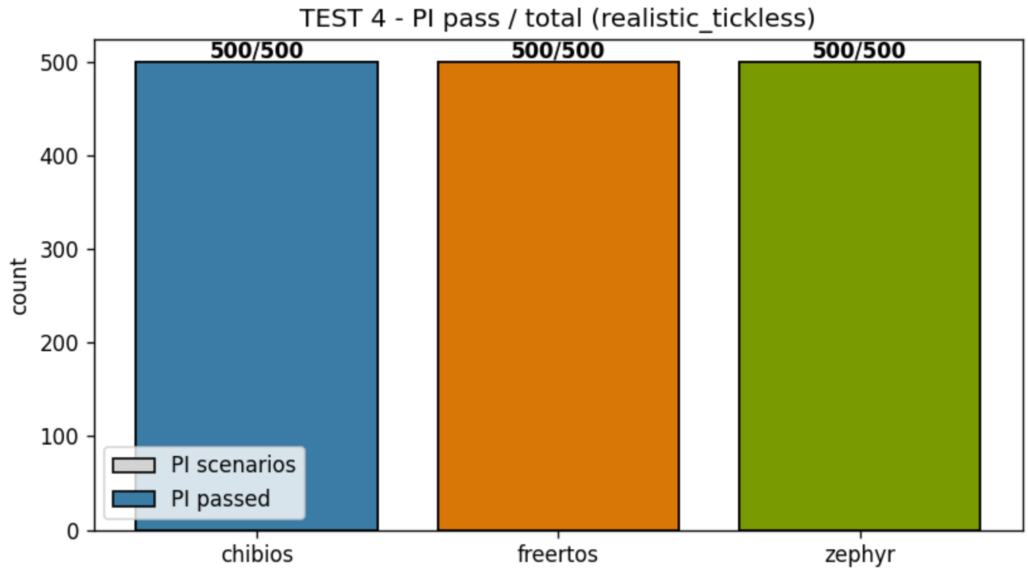


Aggregate distribution - realistic_tickless / t4_mtx_pi



Per-run medians - realistic_tickless / t4_mtx_pi

Priority-inheritance event distribution



T4 priority-inheritance scenarios - realistic_tickless

Cross-profile delta (fair_perf -> realistic_tickless)

Per-cell difference of the median latency between the two publishable profiles. Positive numbers mean realistic_tickless is slower than fair_perf. The largest expected swing is on T1 for FreeRTOS, where enabling the tickless idle path adds a re-arm overhead to every IRQ wake-up.

RTOS	Test	fair_perf median (cyc)	realistic_tickless median (cyc)	delta (cyc)	delta (us)
ChibiOS	t1_irq	298	305	+7	+0.015
ChibiOS	t2_handoff	97	93	-4	-0.008
ChibiOS	t3_mtx_uncont	62	62	+0	+0.000
ChibiOS	t4_mtx_pi	207	204	-3	-0.006
FreeRTOS	t1_irq	443	845	**+402**	**+0.838**
FreeRTOS	t2_handoff	372	381	+9	+0.019
FreeRTOS	t3_mtx_uncont	207	195	-12	-0.025
FreeRTOS	t4_mtx_pi	603	613	+10	+0.021
Zephyr	t1_irq	589	581	-8	-0.017
Zephyr	t2_handoff	424	429	+5	+0.010
Zephyr	t3_mtx_uncont	124	125	+1	+0.002
Zephyr	t4_mtx_pi	721	682	-39	-0.081

Integrity and reproducibility

The publication gate is enforced at three layers: (1) per-run validation, captured in the **.validated.json* manifest written by the collector alongside the CSV; (2) per-campaign lock, written by the lab campaign script after every (RTOS, profile) is complete; (3) build reproducibility, expressed as ELF / MAP SHA-256 homogeneity across the five runs of a given target. All numbers in this report come exclusively from validated runs.

Campaign locks - ELF and MAP SHA-256

fair_perf

RTOS	Artefact	SHA-256 (truncated)
ChibiOS	ELF	`e1223ca085e005ee7fd2bdd3ebb36e63d431be1df4fc582b...`
	MAP	`cb4f7c4325a7e6c824f897e24f45e5d18745483017f677da...`
FreeRTOS	ELF	`1106346d576667e9512b874a38b5e9bf4b298230a9e4bd4b...`
	MAP	`a091586ab780369e74f0b78fc6a94672287c96db61b38692...`
Zephyr	ELF	`e49e45b3d8b6fd7f50b3e21fb136d9a6c6b61fbc400e6563...`
	MAP	`168754812d67b36d150146a94622b31ca3cf543fed6b5667...`

realistic_tickless

RTOS	Artefact	SHA-256 (truncated)
ChibiOS	ELF	`e73fe8eb4a9750364af315168c60923fc79e9c6d3d0f0395...`
	MAP	`72f8d1354d936af4eb1449c90d027ca6b76b49aee5e1222e...`
FreeRTOS	ELF	`cbeb5ddcaa168a9810469c73331413b251e1ddd5710caad5...`
	MAP	`6fad7c42d1f316d96b18eb2506277bf9de0d1da70e42210e...`
Zephyr	ELF	`60785c90379b036adcfaaeefeab965afcb46d92170841967...`
	MAP	`5a46fd04abeecfbad23fbf4e7b4a80a9571669b6afd75848...`

Per-run validation status

RTOS	Profile	Run	Validated	clock (Hz)	VOS	FLASH_ACR	tickless	WFI
ChibiOS	fair_perf	01	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	fair_perf	02	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	fair_perf	03	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	fair_perf	04	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	fair_perf	05	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	01	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	02	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	03	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	04	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	05	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	01	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	02	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	03	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	04	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	05	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	realistic_tickless	01	yes	480000000	VOS0	0x00000034	ON	ON

ChibiOS	realistic_tickless	02	yes	480000000	VOS0	0x00000034	ON	ON
ChibiOS	realistic_tickless	03	yes	480000000	VOS0	0x00000034	ON	ON
ChibiOS	realistic_tickless	04	yes	480000000	VOS0	0x00000034	ON	ON
ChibiOS	realistic_tickless	05	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	01	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	02	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	03	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	04	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	05	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	01	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	02	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	03	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	04	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	05	yes	480000000	VOS0	0x00000034	ON	ON

Conclusions

• **T1 - IRQ -> thread:** In these tests, under the tested configuration, ChibiOS showed the lowest median latency in both profiles. fair_perf medians: ChibiOS 298, FreeRTOS 443, Zephyr 589 cycles. Under realistic_tickless FreeRTOS median rises to 845 cycles, while ChibiOS and Zephyr remain within a few cycles of their fair_perf number.

• **T2 - Thread handoff:** ChibiOS showed the lowest median latency at 97 cycles, against FreeRTOS 372 and Zephyr 424. The measured median ratio in this test is approximately 4x, attributable to the simpler scheduler path on equal-priority yield.

• **T3 - Mutex uncontended:** ChibiOS 62 cycles is the shortest fast path; Zephyr 124 is second. FreeRTOS 207 uses queue-based semaphore primitives in this port; the measured uncontended mutex path is therefore higher in this benchmark.

• **T4 - Mutex contended + PI:** ChibiOS showed the lowest contended-path latency at 207 cycles, with FreeRTOS at 603 and Zephyr at 721. Priority inheritance is correct in all three kernels: 500/500 PI scenarios pass per RTOS per profile.

• **Reproducibility:** run_spread is zero cycles on every (RTOS, test) cell across the campaign. All 30 published run01..run05 captures (3 RTOS x 2 profiles x 5 firmware loads) passed the publication gate; the six run00 warmup captures are discarded.

• **Honest scope:** Phase 1 publishes the DWT cycle delta A4 - A1 for T1, not the hardware-event-to-thread external latency. The hardware-routed IRQ entry path (NVIC, stacking, vector fetch) is intentionally excluded and will be measured by the logic-analyzer in Phase 2 per ADR-015.

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This document is not legal advice; a final legal review is recommended before official publication.

FreeRTOS and Zephyr are excellent RTOS projects with different design goals. This comparison focuses only on real-time latency under the tested conditions.

Appendix A - Build commands

Each port has a single, deterministic build entry point. All ports honour the *PROFILE* selector (fair_perf, realistic_tickless, debug_dev) so that the same source tree produces the publication firmware variants without ambiguity.

ChibiOS

```
cd chibios/benchmark_chibios
make PROFILE=fair_perf -j
make PROFILE=realistic_tickless -j
```

FreeRTOS

```
cd freertos/benchmark_freertos
cmake -B build/fair_perf -G "MinGW Makefiles" -DPROFILE=fair_perf
cmake --build build/fair_perf
cmake -B build/realistic_tickless -G "MinGW Makefiles" -DPROFILE=realistic_tickless
cmake --build build/realistic_tickless
```

Zephyr (board stm32h750b_dk)

```
cd zephyr
.venv/Scripts/activate.bat
west build -d build/fair_perf -b stm32h750b_dk benchmark_zephyr -p always -- -DPROFILE=fair_perf
west build -d build/realistic_tickless -b stm32h750b_dk benchmark_zephyr -p always --
-DPROFILE=realistic_tickless
```

Appendix B - GPIO mapping (logic-analyzer)

Six GPIO signals are exposed on the STMod+ connector P1 of the STM32H750B-DK for logic-analyzer capture. In Phase 1 they are routed by the firmware for parity with the future Phase 2 dual-source measurement and are not part of the published headline. See ADR-007 for the rationale behind the pin assignment.

Signal	MCU pin	STMod+ P1 pin	Role
A0_HW	PA0	1	TIM2 CH1 PWM mode 2 - hardware IRQ stimulus for T1
A1	PH1	17	LOW_LOCK - mutex lock by LOW (T4)
A2	PH4	19	HIGH_WAIT - HIGH blocked on mutex (T4)
A3	PH8	20	LOW_UNLOCK - LOW releases mutex (T4)
A4	PH12	11	HIGH_ACQUIRE - HIGH acquires mutex (T4)
MEDIUM_RUN	PI11	18	MEDIUM scheduled (PI excludes MEDIUM, T4)